

Key idea

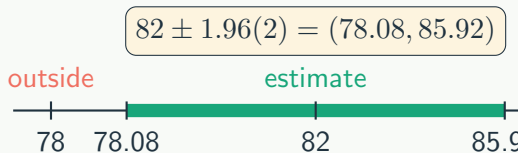
For a Grade 11 learner beginning statistical inference.

Mathematical relationship

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

A 95% confidence interval for a mean with known population standard deviation. Same units as the measured variable

- ▶ Distinguish a sample distribution from a sampling distribution.
- ▶ Interpret standard error as sample-to-sample variability.
- ▶ Compute and interpret a z-score and



Key idea

Random samples vary. The distribution of their sample means has center μ and standard error σ divided by the square root of n .

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- ▶ Distinguish a sample distribution from a sampling distribution.
- ▶ Interpret standard error as sample-to-sample variability.

Sample 1



Sample 2



Sample 3



different samples $\Rightarrow$ different $\bar{x}$

Key idea

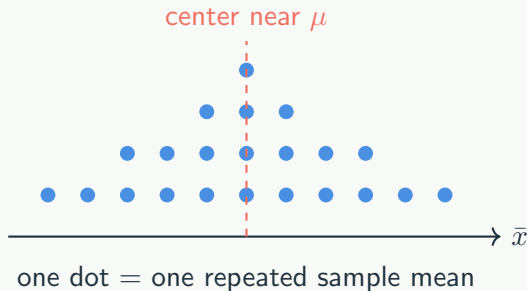
These words unlock the lesson.

Mathematical relationship

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Standard error of the sample mean when population standard deviation is known. Same units as the measured variable

- ▶ population
- ▶ random sample
- ▶ parameter



Key idea

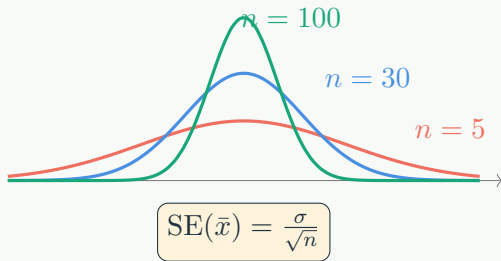
Check the foundations first.

Mathematical relationship

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

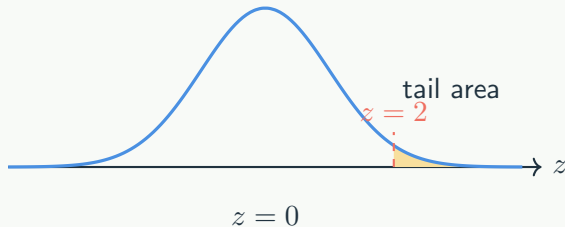
A 95% confidence interval for a mean with known population standard deviation. Same units as the measured variable

- ▶ Find a mean.
- ▶ Evaluate a square root.
- ▶ Read an interval on a number line.



Question

Two random samples from the same population have different means. Is either sample necessarily wrong?



Key Idea 1

Σ Solve

Key idea

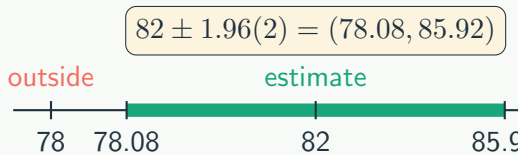
Understand the idea before memorizing details.

Random samples vary. The distribution of their sample means has center μ and standard error σ divided by the square root of n .

Mathematical relationship

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

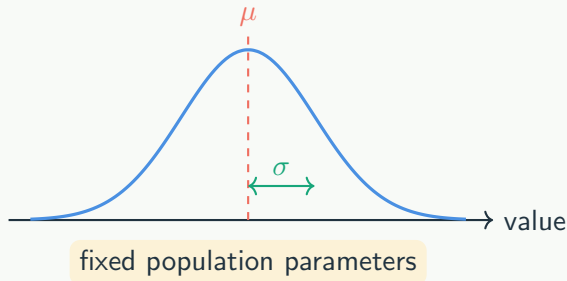
Standard error of the sample mean when population standard deviation is known. Same units as the measured variable



Key idea

A population has fixed parameters. A random sample produces statistics that vary from sample to sample.

- ▶ Population: the full process or group.
- ▶ Sample: observations selected by a random mechanism.



Why Sample Means Vary

Σ Solve

Key idea

Different random samples usually contain different observations, so their sample means are not identical.

Sample 1 — 

Sample 2 — 

Sample 3 — 

different samples $\Rightarrow$ different $\bar{x}$

Build A Sampling Distribution

Σ Solve

Key idea

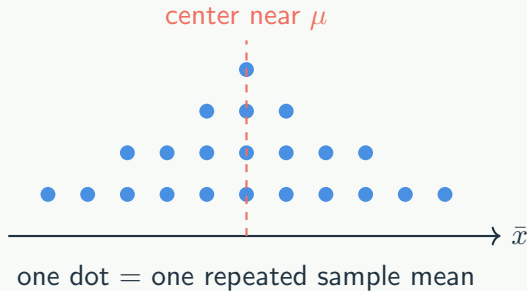
Collecting one sample mean from each repeated sample creates the sampling distribution of the mean.

Mathematical relationship

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Standard error of the sample mean when population standard deviation is known. Same units as the measured variable

- ▶ One dot represents one sample mean.
- ▶ The center is near the population mean.



Sample Size Changes Precision

Key idea

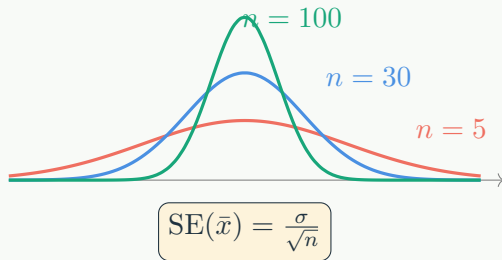
For fixed population spread, larger samples produce a narrower sampling distribution.

Mathematical relationship

$$\widehat{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Estimated standard error of the sample mean.

- ▶ Standard error is the standard deviation of the sampling distribution.
- ▶ Quadrupling n halves the standard error.



Key idea

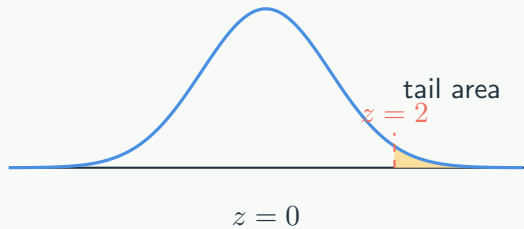
A z-score is a signed distance measured in standard errors. It is not itself a probability.

Mathematical relationship

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

One-sample z statistic for a sample mean.

- ▶ Positive z is above the model mean.
- ▶ Tail area converts location into probability.



Complete z And Interval Example

Σ Solve

Key idea

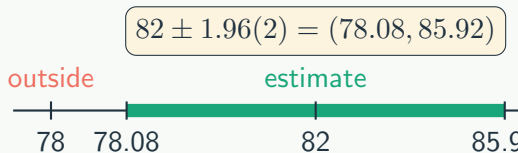
Given $\bar{x}=82$, $\mu=78$, $\sigma=10$, and $n=25$, first compute $SE=2$ and then $z=2$.

Mathematical relationship

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

A 95% confidence interval for a mean with known population standard deviation. Same units as the measured variable

1. Given: $\bar{x}=82$, $\mu=78$, $\sigma=10$, $n=25$.
2. $SE=10/\sqrt{25}=2$.
3. $z=(82-78)/2=2$.



Key idea

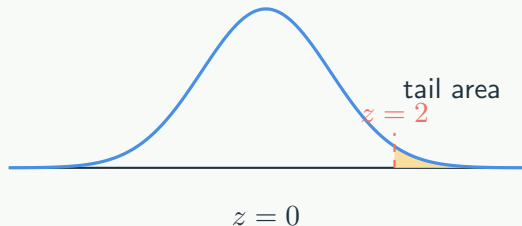
Known population standard deviation supports a z-procedure. When it is unknown, estimate with s and normally use t with n-1 degrees of freedom.

Mathematical relationship

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

One-sample z statistic for a sample mean.

- ▶ Known σ :
 $SE = \sigma / \sqrt{n}$.
- ▶ Unknown σ : estimated
 $SE = s / \sqrt{n}$ and use t.



Example: Given $\bar{x}$ bar 82

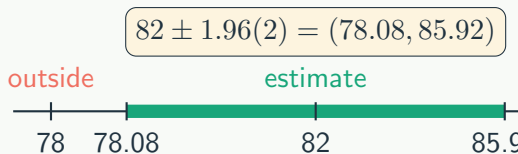
Σ Solve

Given $\bar{x} = 82$, $\mu = 78$, $\sigma = 10$, and $n = 25$. Standard error is 2, z is 2, and the 95% confidence interval is (78.08, 85.92). Therefore 78 is slightly outside the interval.

Mathematical relationship

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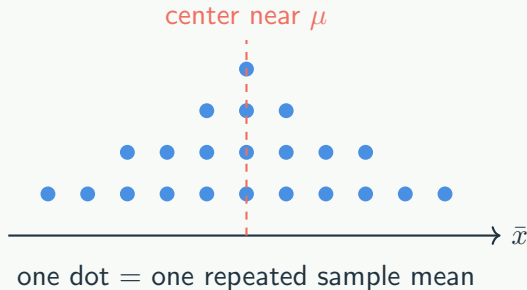
Question

Known sigma is 12 and n is 36. Find the standard error without revealing the answer first.

Mathematical relationship

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Standard error of the sample mean when population standard deviation is known. Same units as the measured variable



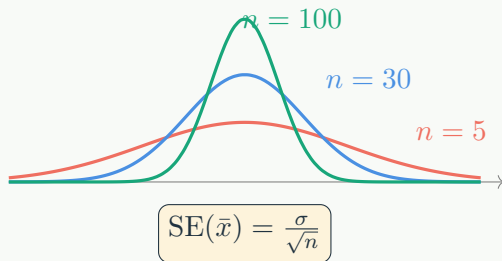
Question

A sample mean is 50, mu is 45, sigma is 8, and n is 16. Find and interpret z.

Mathematical relationship

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

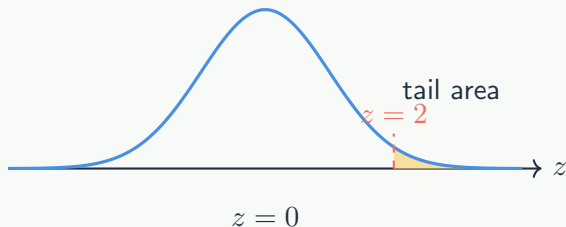
One-sample z statistic for a sample mean.



Key idea

Mistakes become easier to catch when you know what to look for.

- ▶ Do not multiply only one quantity in a ratio.
- ▶ Equivalent ratios need the same scale factor on both quantities.



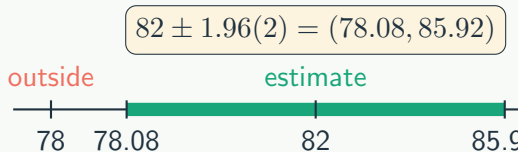
Question

What does one dot in a sampling distribution represent?

Mathematical relationship

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Standard error of the sample mean when population standard deviation is known. Same units as the measured variable



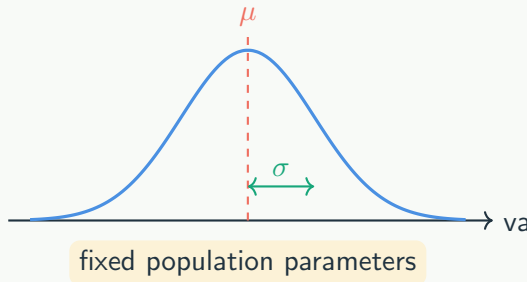
Question

Why does quadrupling n halve standard error?

Mathematical relationship

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Standard error of the sample mean when population standard deviation is known. Same units as the measured variable



Question

When sigma is unknown, why is a t-procedure normally used?

Mathematical relationship

$$\widehat{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Estimated standard error of the sample mean.

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Key idea

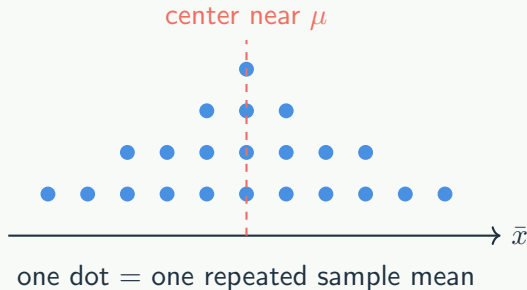
A strong finish means you can explain the idea in your own words.

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Key idea

Use simulation to connect z-scores with one-sided and two-sided probabilities.

Question

What should we practice next?

Mathematical relationship

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

One-sample z statistic for a sample mean.

